

Sainava Modak

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SUMMARY

Computer Science undergraduate with strong foundations in data structures, algorithms, and software engineering. Experience covers building scalable, event-driven systems and applying modern development practices using Java, Python, and Node.js, with a focus on writing secure, efficient, and maintainable code in collaborative environments.

EDUCATION

Amity University, Noida <i>B.Tech, Computer Science and Engineering; CGPA: 9.51; 100% Merit Scholarship</i>	2023 – 2027
Army Public School, Shillong <i>CBSE Senior Secondary (Class XII); Percentage: 97%</i>	2020 – 2022

TECHNICAL SKILLS

Programming: Java, Python, C++, JavaScript
Core Concepts: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems
Backend & Databases: Spring Boot, Node.js, Express, Kafka, REST, MySQL, MongoDB, Redis
Machine Learning: PyTorch, OpenCV, CNN, LSTM, NumPy, Pandas
Cloud & Tools: AWS (Cloud Practitioner), Docker, CI/CD, Git, GitHub, Postman

PROJECTS

CV-Graph-RAG: Structural Literature Synthesis Python, Neo4j, Llama 3, Flask	GitHub
- Engineered a hybrid GraphRAG pipeline fusing a Neo4j knowledge graph with vector search, eliminating cross-document hallucinations and establishing deterministic multi-hop relational integrity across literature. - Built an end-to-end processing pipeline featuring structured entity resolution, localized inference via Ollama, and an interactive dashboard displaying explicit, citation-backed evidence paths.	
LingoStream Java, Spring Boot, Redis, PostgreSQL, WebSockets	GitHub
- Built highly concurrent distributed backend using Spring Boot and Redisson distributed locks, implementing a Redis cache-aside pattern to prevent data collisions and reduce read latency to under 5ms. - Engineered real-time collaborative editing using STOMP WebSockets for full-duplex broadcasting, securing the Docker-containerized API architecture with stateless JWT authentication.	
Tactical GNN Spatial Engine Python, PyTorch, React, Graph Neural Networks	GitHub
- Engineered a spatial sports analytics engine using PyTorch and Graph Attention Networks (GATv2) to process raw tracking data and model dynamic spatial dominance. - Built an interactive React dashboard utilizing Voronoi polygons and custom kinematic vector visualizations to map real-time player momentum, acceleration, and pitch control.	

INTERNATIONAL EXPERIENCE & OPEN SOURCE

Talent for Japan 2025 Winner (Selected among Top 11 from 5000+ applicants): Represented India in Japan; presented AI-driven Smart Factory Intelligence solutions to Japanese companies and media. Open Source: Ranked 14th in OSCI 2025 and 33rd in SSOC 2025; contributed to GSSOC and Hacktoberfest.
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INTERNSHIPS

Conversia <i>Full-Stack Development Intern</i>	Jan 2025 – Apr 2025
- Optimized Node.js/Express REST APIs for an AI-driven platform, reducing response time by 20% and improving system throughput. - Developed scalable backend services and API endpoints supporting real-time AI workflows, enhancing system reliability and maintainability.	

RESEARCH

Multimodal Deepfake Detection (Ongoing) Python, PyTorch, OpenCV, Bi-LSTM	GitHub
- Built multimodal deepfake detection pipeline using RGB, optical flow, and Bi-LSTM. - Achieved 97% accuracy on FaceForensics++ and 95%+ on CelebDF datasets.	

CERTIFICATIONS

Oracle Cloud Infrastructure 2025 Certified Generative AI Professional AWS Certified Cloud Practitioner Python for Data Science (NPTEL) — Top 5%
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